

Quiz study guide: linear systems, distances & RANSAC

Predicted questions with correct answers and the theory behind each one

Inhomogeneous systems

Homogeneous systems

Point-line / point-plane distance

RANSAC & minimal samples

Built from the course materials: matrix.pdf, Lagrange-1.pdf, CircleEllipseFitting.pdf, the RANSAC lab notebook (Lab 3), and last week's 5 quiz questions.

How the quiz is scored

5 questions at the start of the lab. 4–5 correct: **+1**. 3 correct: **0**. Fewer: **-1**.

What last week's quiz tells us

- All 5 questions were **formula recognition**, with no calculations and no derivations.
- Every answer was a formula taken **directly from the PDF** (matrix.pdf §2.1.1 → §3, in order).
- The wrong options were **near-misses built from the same symbols** ($2x$, x , $2Ax$, a).
- 3 of the 5** were variations of one idea, so expect several "how many points for model X?" questions.
- Circle fitting was announced but **not asked**, so the quiz favors the headline formulas.

The 5 most likely questions

- Minimal points** for a line / plane / circle / sphere ($2 / 3 / 3 / 4$). **VERY HIGH**
- Homogeneous** $Ax = 0$: the eigenvector of $A^T A$ with the **smallest** eigenvalue. **VERY HIGH**
- Distance** to $\mathbf{l}^T \mathbf{x} + m = 0$: $|\mathbf{l}^T \mathbf{x}_0 + m| / \sqrt{\mathbf{l}^T \mathbf{l}}$. **HIGH**
- Inhomogeneous** $Ax = b$: $A^T Ax = A^T b$, $x = (A^T A)^{-1} A^T b$. **HIGH**
- RANSAC iterations**: $\log(1 - p) / \log(1 - w^s)$. **MEDIUM**

Forecast at a glance

Tier	Covers	Questions
VERY HIGH	Minimal points (line, plane, sphere, circle); solution of the homogeneous system	Q1-Q5
HIGH	Distance formulas (general, unit-normal, plane, 2D line); inhomogeneous solution and normal equation; the $y = mx + b$ system; ellipse points	Q6-Q13
MEDIUM	RANSAC iteration formula and w^s ; constraint, Lagrangian and eigen-equation; SVD; centroid and normal of the optimal plane; what k doesn't depend on; gradient of $\ Ax - b\ ^2$; homogeneous vs. inhomogeneous; 3D line; cost = $\lambda_{\min}$; the $z = ax + by + c$ system	Q14-Q27
BONUS	Numeric versions and RANSAC concept questions (less likely given the format)	Q28-Q33

Contents

Part 1 • Theory (§1 gradients • §2 inhomogeneous • §3 Lagrange • §4 homogeneous & SVD • §5 distances • §6 optimal fitting • §7 RANSAC • §8 minimal samples • §9 traps)

Part 2 • 33 predicted questions with answers

Part 3 • One-page cheat sheet

Theory you need

Each section is short and follows the course PDFs. Every question in Part 2 points back to a section here (§).

§1 Gradient rules (recap from last week)

These are needed for every derivation this week. Last week's quiz asked the first four directly.

Scalar function J	$\partial J / \partial x$	Note
$x^T x$	$2x$	squared L2 norm
$a^T x = x^T a$	a	scalar product
$x^T A x$	$(A^T + A)x$	general square A
$x^T B x, B = B^T$	$2Bx$	symmetric case (e.g. $B = A^T A$)
$\ Ax - b\ ^2$	$2A^T Ax - 2A^T b$	inhomogeneous least squares
$x^T A^T A x$	$2A^T A x$	homogeneous least squares

§2 Inhomogeneous overdetermined systems $Ax = b$

Source: matrix.pdf §3-4

A is $m \times n$ with more rows than columns ($m > n$): more equations than unknowns. With noisy data there is no exact solution, so we minimize the squared residual:

$$x^* = \arg \min_x \|Ax - b\|_2^2, \quad J = (Ax - b)^T (Ax - b) = x^T A^T A x - 2x^T A^T b + b^T b.$$

matrix.pdf has a typo here: it writes $2x^T A^T x$. The middle term is $-2x^T A^T b$.

Setting the gradient to zero:

$$\frac{\partial J}{\partial x} = 2A^T A x - 2A^T b = 0 \implies \boxed{A^T A x = A^T b} \text{ (normal equation)} \implies \boxed{x = (A^T A)^{-1} A^T b = A^+ b}$$

$A^+ = (A^T A)^{-1} A^T$ is the **Moore-Penrose pseudo-inverse** (for $m > n$, full column rank).

Case studies (matrix.pdf §4, and last week's circle fitting)

Model	Row i of A	Unknowns x	b_i
Line $y = mx + b$	$[x_i \ 1]$	$[m \ b]^T$	y_i
Plane $z = ax + by + c$	$[x_i \ y_i \ 1]$	$[a \ b \ c]^T$	z_i
Circle	$[1 \ -2x_i \ -2y_i]$	$[x_0^2 + y_0^2 - R^2, x_0, y_0]^T$	$-x_i^2 - y_i^2$
Sphere	$[1 \ -2x_i \ -2y_i \ -2z_i]$	$[x_0^2 + y_0^2 + z_0^2 - R^2, x_0, y_0, z_0]^T$	$-x_i^2 - y_i^2 - z_i^2$

For the line, the normal equation in closed form is $\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & N \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}$.

How to recognize "inhomogeneous": the right side is a known, nonzero vector. One coefficient has been fixed to 1 (the coefficient of y , of z , or of $x^2 + y^2$).

Limitations: explicit forms minimize **vertical** residuals (in y or z), not perpendicular distances, and $y = mx + b$ cannot represent a vertical line.

Typo alert: matrix.pdf calls the $y = mx + b$ system "homogeneous". It is **inhomogeneous**.

§3 Lagrange multipliers

Source: Lagrange-1.pdf §1

Goal: minimize $f(\mathbf{x})$ subject to $g(\mathbf{x}) = c$. At the optimum the level curve of f is **tangent** to the constraint curve, so their gradients (normals) are parallel: $\nabla f = \lambda \nabla g$. Lagrange packed both conditions into one cost function:

$$J(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c)$$

- $\partial J / \partial \mathbf{x} = 0 \Rightarrow \nabla f = \lambda \nabla g$ (parallel gradients)
- $\partial J / \partial \lambda = 0 \Rightarrow g(\mathbf{x}) = c$ (gives back the constraint)

λ is the **Lagrange multiplier**. The method works in any dimension.

§4 Homogeneous overdetermined systems $Ax = 0$

Source: Lagrange-1.pdf §2, §3, §5

Two problems: $x = 0$ always solves it (the trivial solution), and if x is a solution so is αx (the scale is free). So we fix the length with the constraint $x^T x = 1$ and minimize $\|Ax\|^2$:

$$J = x^T A^T A x - \lambda(x^T x - 1), \quad \frac{\partial J}{\partial x} = 2A^T A x - 2\lambda x = 0 \Rightarrow A^T A x = \lambda x$$

This is an **eigenvalue problem**. Substituting back: $x^T A^T A x = \lambda x^T x = \lambda$, so the cost equals the eigenvalue.

Solution: the eigenvector of $A^T A$ corresponding to the **smallest eigenvalue**. (For maximizing $\|Ax\|$, take the largest.)

$A^T A$ is symmetric positive semidefinite, so its eigenvalues are **real and non-negative**.

Same result using the SVD

Every matrix has an SVD $A = U S V^T$, where U and V are orthogonal ($U^T U = I$, $V^T V = I$) and S is diagonal with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

- The columns of V are the eigenvectors of $A^T A$. The columns of U are the eigenvectors of $A A^T$.
- $\sigma_i = \sqrt{\lambda_i}$. The eigenvalues of $A A^T$ and $A^T A$ are the same and are listed in descending order.
- So the solution of $Ax = 0$ is the **last column of V** , the right singular vector of the smallest singular value.

Examples

- **Implicit plane** $ax + by + cz + d = 0$: rows $[x_i \ y_i \ z_i \ 1]$, unknowns $[a \ b \ c \ d]^T$. The solution is the eigenvector with the smallest eigenvalue of the 4×4 matrix of sums $\begin{bmatrix} \sum x_i^2 & \sum x_i y_i & \sum x_i z_i & \sum x_i \\ \dots & \dots & \dots & \dots \\ \sum x_i & \sum y_i & \sum z_i & N \end{bmatrix}$.
- **Ellipse** (last week): $Mz = 0$ with 6 unknowns and the extra constraint $z^T K z = -1$ (so that $B^2 - 4AC < 0$). This gives the generalized eigenproblem $M^T M z = -\lambda K z$.

Counting: n unknowns but only $n - 1$ degrees of freedom (the scale is free), so you need at least $n - 1$ independent equations.

	Inhomogeneous $Ax = b$	Homogeneous $Ax = 0$
Right side	known, nonzero b	zero
Minimize	$\ Ax - b\ ^2$	$\ Ax\ ^2$ s.t. $x^T x = 1$
Key equation	$A^T A x = A^T b$	$A^T A x = \lambda x$
Solution	$(A^T A)^{-1} A^T b$	eigenvector of $A^T A$, smallest λ = last column of V
Course examples	$y = mx + b$, $z = ax + by + c$, circle, sphere	implicit line/plane, ellipse

§5 Point-line and point-plane distance

Source: Lagrange-1.pdf §4.1

Write any line/plane/hyperplane as $\mathbf{l}^T \mathbf{x} + m = 0$, where $\mathbf{l} = [a \ b]^T$ for a line and $\mathbf{l} = [a \ b \ c]^T$ for a plane. $\mathbf{l}$ is the **normal vector**. To find the closest point $\mathbf{x}$ to $\mathbf{x}_0$, minimize the squared distance subject to $\mathbf{x}$ lying on the hyperplane:

$$J = (\mathbf{x} - \mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) + \lambda (\mathbf{l}^T \mathbf{x} + m) \Rightarrow 2(\mathbf{x} - \mathbf{x}_0) + \lambda \mathbf{l} = 0 \Rightarrow \mathbf{x} = \mathbf{x}_0 - \frac{\lambda}{2} \mathbf{l}$$

Substituting into the constraint gives $\lambda = 2 \frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}}$, so $\mathbf{x} - \mathbf{x}_0 = -\frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}} \mathbf{l}$ and

$$d = \frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\sqrt{\mathbf{l}^T \mathbf{l}}} \quad \text{and if } \mathbf{l}^T \mathbf{l} = 1: \quad d = |\mathbf{l}^T \mathbf{x}_0 + m|$$

Case	Distance	Example
2D line $ax + by + c = 0$	$\frac{ ax_0 + by_0 + c }{\sqrt{a^2 + b^2}}$	$(1, 2), 3x + 4y - 1 = 0: 10/5 = 2$
Plane $ax + by + cz + d = 0$	$\frac{ ax_0 + by_0 + cz_0 + d }{\sqrt{a^2 + b^2 + c^2}}$	$(1, 1, 1), 2x + y + 2z + 4 = 0: 9/3 = 3$
Circle / sphere (center $\mathbf{c}$, radius R)	$ \ \mathbf{p} - \mathbf{c}\ - R $	used for RANSAC inlier tests

- The constant term (m , c , or d) is **never under the square root**. Only the normal's components are.
- Without the absolute value you get the **signed** distance, and its sign tells you which side of the line/plane the point is on. The formula is valid in any dimension.
- The RANSAC lab builds the normal as $(a, b) = (-\Delta y, \Delta x) / \|\Delta\|$, so $\sqrt{a^2 + b^2} = 1$ and the distance is just $\text{abs}(a*x + b*y + c)$.

Lagrange-1.pdf writes $\mathbf{l}^T \mathbf{x} + m$ in the final numerator where it means $\mathbf{l}^T \mathbf{x}_0 + m$.

§6 Optimal line/plane fitting (perpendicular distances)

Source: Lagrange-1.pdf §4.2-4.3

Minimize the sum of squared **geometric** distances $\sum_i d_i^2 = \sum_i (\mathbf{l}^T \mathbf{x}_i + m)^2$ with $\mathbf{l}^T \mathbf{l} = 1$. This splits into two steps:

- Translation.** $\partial J / \partial t = 0 \Rightarrow t = -\frac{1}{N} \sum \mathbf{x}_i$. The optimal line/plane passes through the **center of gravity** $\bar{\mathbf{x}}$.
- Rotation.** Center the points and stack them as the rows of X . Then solve $\min \sum (\mathbf{l}^T \mathbf{x}_i)^2$ s.t. $\mathbf{l}^T \mathbf{l} = 1$, which is the homogeneous problem $X \mathbf{l} = 0$. The **normal** $\mathbf{l}$ is the eigenvector of $X^T X$ with the **smallest** eigenvalue. Finally $m = -\mathbf{l}^T \bar{\mathbf{x}}$.

For a 2D line, the line *direction* is the eigenvector with the **largest** eigenvalue (this is PCA). The smallest eigenvalue equals the residual $\sum d_i^2$. RANSAC uses this procedure for its final refit.

§7 RANSAC (RANDOM SAMPLE CONSENSUS)

Source: Lab 3 notebook

Why: least squares uses a *squared* cost, so a single far outlier can ruin the fit. RANSAC finds the model supported by the **most points**.

- Randomly pick a **minimal sample** of s points (for a line, $s = 2$).
- Fit the model to the sample. Skip degenerate samples, e.g. coincident points (the lab checks $\|\Delta\| < 10^{-8}$).
- Compute the distance of **every** point to the model.
- Points closer than the threshold are **inliers**. The lab uses $\sigma = 10$ px, chosen from the noise level.
- If this model has more inliers than the best so far, store it and **update the iteration count** using $w = \text{inliers} / N$.
- Stop after k iterations, then **refit** the model with least squares on all inliers (§6).

Number of iterations

$P(\text{sample is all inliers}) = w^s$, and $P(\text{all } k \text{ samples are bad}) = (1 - w^s)^k$. Requiring this to be at most $1 - p$:

$$k = \left\lceil \frac{\log(1-p)}{\log(1-w^s)} \right\rceil$$

p = confidence, w = inlier ratio, s = sample size

w	$p = 0.99$				$p = 0.95$			
	$s = 2$	3	4	5	$s = 2$	3	4	5
0.8	5	7	9	12	3	5	6	8
0.7	7	11	17	26	5	8	11	17
0.5	17	35	72	146	11	23	47	95
0.3	49	169	567	1893	32	110	369	1232

- k depends only on p , w and s , not on N directly. It grows exponentially with s , which is why samples are minimal.
- Adaptive:** w is unknown at the start, so it is estimated from the best model found so far, and k shrinks as better models appear (the lab printed $256 \rightarrow 125 \rightarrow 10$ with $p = 0.95$, $w \approx 0.52$).
- RANSAC is **non-deterministic**. To fit several models (e.g. several planes), run it **sequentially**: find a model, remove its inliers, repeat.

§8 Minimal sample sizes

Rule: $s = \lceil \text{DoF} / \text{constraints per point} \rceil$. A point on a 2D curve or a 3D surface gives **1** equation. A point on a 3D curve (3D line, circle in 3D) gives **2**. For homogeneous (implicit) models, DoF = number of coefficients - 1.

Model	Equation	DoF	Min. points	Degenerate if...
2D line	$ax + by + c = 0$	2	2	points coincide
Plane	$ax + by + cz + d = 0$	3	3	points collinear
2D circle	$(x - x_0)^2 + (y - y_0)^2 = R^2$	3	3	points collinear
Sphere	$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2$	4	4	points coplanar
Ellipse / conic	$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$	5	5	—
Parabola	$y = ax^2 + bx + c$	3	3	—
3D line	point + direction	4	2	points coincide
Circle in 3D	center, normal, radius	6	3	points collinear
Cylinder	axis (4) + radius	5	5 (2 if normals known)	—

§9 Near-miss traps (how last week's wrong answers were built)

Last week every wrong option was a close variant of the right one (e.g. $2Ax$ vs. $(A^T + A)x$). Watch for:

Smallest vs. largest eigenvalue

$A^T A$ vs. AA^T ; columns of V vs. U ; **last** vs. first column

$\sqrt{I^T I}$ vs. $I^T I$ vs. no denominator

The constant (m, c, d) must **not** go under the square root

$\log(1 - p) / \log(1 - w^s)$ vs. $\log p / \log w^s$

w^s vs. $1 - w^s$ vs. $(1 - w)^s$

Sphere: **4**, not 3. Ellipse: **5**, not 6. Plane: **3**, not 4.

$A^T A x = A^T b$ (inhomogeneous) vs. $A^T A x = \lambda x$ (homogeneous)

Predicted questions, with answers

Grouped by likelihood and written in last week's format: a short question, four formula options, and near-miss distractors. The correct option is marked in green. Tip: cover the green line and the explanation, answer, then check.

Very high Q1-Q5 · 5 questions

Most likely to appear. Learn these first.

Q1 VERY HIGH RANSAC · minimal sample

How many points are required to estimate a **2D line** minimally?

A 1

B 2 ✓

C 3

D 4

Answer B The implicit line $ax + by + c = 0$ has 3 coefficients but is defined only up to scale, so it has **2 degrees of freedom**. Each point on the line gives one equation, so 2 points are enough. This is exactly what the lab's `fit_line_two_points` does. The two points must not coincide. [Theory §8](#)

Q2 VERY HIGH RANSAC · minimal sample

How many points are required to estimate a **plane** minimally?

A 2

B 3 ✓

C 4

D 5

Answer B $ax + by + cz + d = 0$ has 4 coefficients, but the scale is free, so there are **3 DoF** and **3 points** are needed. They must **not be collinear**: infinitely many planes pass through a line. Trap: 4 is the number of *unknowns*, not the DoF. [Theory §8](#)

Q3 VERY HIGH RANSAC · minimal sample

How many points are required to estimate a **sphere** minimally?

A 2

B 3

C 4 ✓

D 5

Answer C $(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2$ has **4 unknowns** (x_0, y_0, z_0, R). The linear sphere-fitting system also has 4 columns: $[1 \quad -2x_i \quad -2y_i \quad -2z_i]$. So **4 non-coplanar points** are needed. Trap: 3 is the answer for a circle. [Theory §8](#)

Q4 VERY HIGH RANSAC · minimal sample

How many points are required to estimate a **2D circle** minimally?

A 2

B 3 ✓

C 4

D 5

Answer B A circle has 3 unknowns (x_0, y_0, R). In last week's linear formulation the unknown vector is $[x_0^2 + y_0^2 - R^2, x_0, y_0]^T$, which also has 3 entries. So **3 non-collinear points** are needed. [Theory §8](#)

Q5 **VERY HIGH** Homogeneous systems

What is the optimal (non-trivial) solution of the overdetermined **homogeneous** linear system $Ax = 0$?

- ☐ A The eigenvector of $A^T A$ corresponding to the **largest** eigenvalue
- ☐ B $(A^T A)^{-1} A^T b$
- ☒ C The eigenvector of $A^T A$ corresponding to the **smallest** eigenvalue ✓
- ☐ D The eigenvector of AA^T corresponding to the smallest eigenvalue

Answer C Minimize $x^T A^T A x$ subject to $x^T x = 1$. The Lagrangian gives $A^T A x = \lambda x$, and plugging back in, the cost equals λ . So the minimum is the eigenvector of the **smallest** eigenvalue. The largest eigenvalue gives the *maximum*. AA^T is the wrong matrix, and $(A^T A)^{-1} A^T b$ is the inhomogeneous solution. **Theory §4**

High Q6-Q13 · 8 questions

Likely. The distance and inhomogeneous formulas are the core of this week.

Q6 **HIGH** Point-line/plane distance

What is the distance of the point $\mathbf{x}_0$ from the line/plane $\mathbf{l}^T \mathbf{x} + m = 0$?

- ☐ A $\frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\mathbf{l}^T \mathbf{l}}$
- ☒ B $\frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\sqrt{\mathbf{l}^T \mathbf{l}}}$ ✓
- ☐ C $(\mathbf{l}^T \mathbf{x}_0 + m)^2$
- ☐ D $\sqrt{\mathbf{l}^T \mathbf{x}_0 + m}$

Answer B The Lagrange derivation gives the foot point $\mathbf{x} - \mathbf{x}_0 = -\frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}} \mathbf{l}$. Its length is $\frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\mathbf{l}^T \mathbf{l}} \sqrt{\mathbf{l}^T \mathbf{l}} = \frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\sqrt{\mathbf{l}^T \mathbf{l}}}$. This holds in any dimension. Trap: dividing by $\mathbf{l}^T \mathbf{l}$ without the square root. **Theory §5**

Q7 **HIGH** Point-line/plane distance

What is the distance of $\mathbf{x}_0$ from $\mathbf{l}^T \mathbf{x} + m = 0$ if $\mathbf{l}^T \mathbf{l} = 1$?

- ☒ A $|\mathbf{l}^T \mathbf{x}_0 + m|$ ✓
- ☐ B $(\mathbf{l}^T \mathbf{x}_0 + m)^2$
- ☐ C $|\mathbf{l}^T \mathbf{x}_0 + m|/m$
- ☐ D $\mathbf{l}^T \mathbf{x}_0$

Answer A With a unit normal the denominator $\sqrt{\mathbf{l}^T \mathbf{l}} = 1$, so the **implicit equation gives the distance directly**. That's why the RANSAC lab normalizes (a, b) and computes the distance as $\text{abs}(a*x + b*y + c)$. **Theory §5**

Q8 HIGH Point-line/plane distance

What is the distance of the point $[x_0 \ y_0 \ z_0]^T$ from the plane $ax + by + cz + d = 0$?

☐ A $\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2 + d^2}}$

☒ B $\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$ ✓

☐ C $\frac{|ax_0 + by_0 + cz_0 + d|}{a^2 + b^2 + c^2}$

☐ D $|ax_0 + by_0 + cz_0|$

Answer B Here $\mathbf{l} = [a \ b \ c]^T$ is the normal and $m = d$. Only the normal goes into the norm, so d is **never under the square root**. **Theory §5**

Q9 HIGH Point-line/plane distance

What is the distance of the point $[x_0 \ y_0]^T$ from the line $ax + by + c = 0$?

☐ A $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2 + c^2}}$

☐ B $\frac{|ax_0 + by_0 + c|}{a^2 + b^2}$

☒ C $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$ ✓

☐ D $\sqrt{|ax_0 + by_0 + c|}$

Answer C This is the 2D case of the general formula, with $\mathbf{l} = [a \ b]^T$ and $m = c$. The normal vector is (a, b) . Example: the point $(1, 2)$ and $3x + 4y - 1 = 0$ give $|3 + 8 - 1|/5 = 2$. **Theory §5**

Q10 HIGH Inhomogeneous systems

What is the optimal solution of the overdetermined **inhomogeneous** linear system $Ax = b$?

☐ A $A^{-1}b$

☐ B $A^T b$

☒ C $(A^T A)^{-1} A^T b$ ✓

☐ D The eigenvector of $A^T A$ corresponding to the smallest eigenvalue

Answer C The least-squares optimum of $\|Ax - b\|^2$ is $x = (A^T A)^{-1} A^T b = A^+ b$, where A^+ is the Moore-Penrose pseudo-inverse (asked last week). A is not square, so A^{-1} doesn't exist. The eigenvector answer belongs to the *homogeneous* case. **Theory §2**

Q11 HIGH Inhomogeneous systems

What is the **normal equation** of the overdetermined linear system $Ax = b$?

☒ A $A^T Ax = A^T b$ ✓

☐ B $AA^T x = b$

☐ C $A^T Ax = \lambda x$

☐ D $Ax = 0$

Answer A Setting the gradient of $J = \|Ax - b\|^2$ to zero gives $2A^T Ax - 2A^T b = 0$, which is $A^T Ax = A^T b$. Don't confuse it with $A^T Ax = \lambda x$, the eigenvalue problem from the homogeneous case. **Theory §2**

Q12 HIGH Inhomogeneous systems · line fitting

When fitting the line $y = mx + b$ to points (x_i, y_i) by least squares, what is the i -th row of the coefficient matrix (unknowns $[m \ b]^T$, right side y_i)?

☒ A $[x_i \ 1]$ ✓

☐ B $[x_i \ y_i \ 1]$

☐ C $[y_i \ 1]$

☐ D $[1 \ -2x_i \ -2y_i]$

Answer A Each point gives $y_i = mx_i + b$, so the row is $[x_i \ 1]$ and the right side is y_i . This is matrix.pdf §4.1, the section right after last week's pseudo-inverse question. $[x_i \ y_i \ 1]$ belongs to the implicit homogeneous line, and $[1 \ -2x_i \ -2y_i]$ to circle fitting. **Theory §2**

Q13 HIGH RANSAC · minimal sample

How many points are required to estimate an **ellipse** (general conic $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$) minimally?

☐ A 3

☐ B 4

☒ C 5 ✓

☐ D 6

Answer C There are 6 coefficients, but the conic is **homogeneous** (defined up to scale), so there are **5 DoF** and 5 points are needed. Trap: 6 counts unknowns, not DoF. **Theory §8**

Medium Q14-Q27 · 14 questions

Could be swapped in. Same formulas seen from a different angle.

Q14 MEDIUM RANSAC · iterations

What is the required number of RANSAC iterations? (p : confidence, w : inlier ratio, s : sample size)

☐ A $\frac{\log p}{\log w^s}$

☒ B $\frac{\log(1-p)}{\log(1-w^s)}$ ✓

☐ C $\frac{\log(1-w)}{\log(1-p^s)}$

☐ D $\frac{\log(1-p)}{\log(1-w)^s}$

Answer B A sample is all inliers with probability w^s . All k samples are bad with probability $(1-w^s)^k$. Requiring $(1-w^s)^k \leq 1-p$ and taking logs gives $k \geq \log(1-p)/\log(1-w^s)$, rounded up. **Theory §7**

Q15 MEDIUM RANSAC · probability

What is the probability that a random sample of s points contains **only inliers**? (w : inlier ratio)

☒ A w^s ✓

☐ B $1-w^s$

☐ C $(1-w)^s$

☐ D $s \cdot w$

Answer A Each draw is an inlier with probability w , and there are s independent draws. So $1-w^s$ is the probability that the sample contains at least one outlier, and $(1-w)^s$ is the probability that *all* are outliers. **Theory §7**

Q16 MEDIUM Homogeneous systems

Which constraint is applied when solving the homogeneous system $Ax = 0$?

- ☒ A $x^T x = 1$ ✓ ☐ B $x = 0$ ☐ C $x^T Ax = 0$ ☐ D $Ax = b$

Answer A $x = 0$ always solves $Ax = 0$, and any multiple αx of a solution is also a solution. Requiring **unit length** rules out the trivial solution and fixes the scale. **Theory §4**

Q17 MEDIUM Lagrange multipliers

What is the cost function of the homogeneous problem when using a Lagrange multiplier?

- ☒ A $x^T A^T Ax - \lambda(x^T x - 1)$ ✓ ☐ B $(Ax - b)^T (Ax - b)$
☐ C $x^T x - \lambda(x^T A^T Ax - 1)$ ☐ D $x^T A^T Ax$

Answer A The objective $\|Ax\|^2 = x^T A^T Ax$ minus λ times the constraint $g(x) - c = x^T x - 1$. This follows the general form $J = f(x) - \lambda(g(x) - c)$. **Theory §3-4**

Q18 MEDIUM Lagrange multipliers

Setting $\partial J / \partial x = 0$ for $J = x^T A^T Ax - \lambda(x^T x - 1)$ leads to:

- ☐ A $A^T Ax = A^T b$ ☒ B $A^T Ax = \lambda x$ ✓ ☐ C $Ax = \lambda x$ ☐ D $AA^T x = \lambda x$

Answer B Using the gradient rules: $\partial(x^T A^T Ax) / \partial x = 2A^T Ax$ (since $A^T A$ is symmetric) and $\partial(x^T x) / \partial x = 2x$. So $2A^T Ax - 2\lambda x = 0$, an **eigenvalue problem** of $A^T A$. **Theory §4**

Q19 MEDIUM SVD

Using the SVD $A = USV^T$ (singular values in descending order), where is the solution of $Ax = 0$, $\|x\| = 1$?

- ☐ A First column of U ☐ B Last column of U
☐ C First column of V ☒ D Last column of V ✓

Answer D The columns of V are the eigenvectors of $A^T A$, and the columns of U are those of AA^T . $\sigma_i = \sqrt{\lambda_i}$ in descending order, so the **last column of V** goes with the smallest eigenvalue. **Theory §4**

Q20 MEDIUM Optimal line/plane fitting

In optimal line/plane fitting (minimizing squared perpendicular distances), which point does the fitted line/plane always pass through?

- ☐ A The origin ☒ B The center of gravity of the points ✓
☐ C The first data point ☐ D The point with the largest error

Answer B $\partial J / \partial t = 0$ gives $\sum(\mathbf{x}_i + t) = 0$, so $t = -\frac{1}{N} \sum \mathbf{x}_i$. The optimal translation moves the **centroid** to the origin. **Theory §6**

Q21 MEDIUM Optimal line/plane fitting

After centering the points (rows of X), what is the **normal vector** of the optimal plane?

- A The eigenvector of $X^T X$ with the largest eigenvalue
- B The eigenvector of $X^T X$ with the smallest eigenvalue** ✓
- C $(X^T X)^{-1} X^T \mathbf{1}$
- D The center of gravity

Answer B Minimizing $\sum (\mathbf{l}^T \mathbf{x}_i)^2$ with $\mathbf{l}^T \mathbf{1} = 1$ is the homogeneous problem $X\mathbf{l} = 0$. The normal is the eigenvector with the smallest eigenvalue. The eigenvector with the largest eigenvalue is the dominant *direction*, which is the direction of the line in 2D. [Theory §6](#)

Q22 MEDIUM RANSAC · iterations

Which quantity does **NOT** directly influence the required number of RANSAC iterations?

- A Inlier ratio
- B Size of the minimal sample
- C Required confidence
- D Total number of points** ✓

Answer D $k = \log(1-p)/\log(1-w^s)$ contains only p , w and s . The total number of points N matters only through the ratio $w = \text{inliers}/N$. [Theory §7](#)

Q23 MEDIUM Gradients (recap)

What is the gradient of $J = \|Ax - b\|^2$ with respect to x ?

- A $2Ax - 2b$
- B $2A^T Ax - 2A^T b$** ✓
- C $A^T Ax - b$
- D $2AA^T x - 2Ab$

Answer B Expand $J = x^T A^T Ax - 2x^T A^T b + b^T b$. $A^T A$ is symmetric, so the first term gives $2A^T Ax$. The second is a scalar product with $a = A^T b$, which gives $-2A^T b$. This combines last week's gradient rules with this week's topic. [Theory §1-2](#)

Q24 MEDIUM Homogeneous vs inhomogeneous

Which fitting problem, as formulated in the course materials, leads to a **homogeneous** linear system?

- ☐ A Line fitting with $y = mx + b$
- ☐ B Plane fitting with $z = ax + by + c$
- ☐ C Circle fitting with unknowns $[x_0^2 + y_0^2 - R^2, x_0, y_0]$
- ☒ D Plane fitting with $ax + by + cz + d = 0$ ✓

Answer D The implicit plane gives $[x_i \ y_i \ z_i \ 1][a \ b \ c \ d]^T = 0$, so the right side is zero. The other three have a known, nonzero right side $(y_i, z_i, -x_i^2 - y_i^2)$, so they are **inhomogeneous**. [Theory §2, §4](#)

Q25 MEDIUM RANSAC · minimal sample

How many points are required to estimate a **3D line** minimally?

- ☐ A 1
- ☒ B 2 ✓
- ☐ C 3
- ☐ D 4

Answer B A 3D line has 4 DoF, but a point lying on it gives 2 constraints (it lies on two planes). So $\lceil 4/2 \rceil = 2$ points are enough, which matches the intuition that two points define a line. [Theory §8](#)

Q26 MEDIUM Homogeneous systems

If x is the optimal unit-length solution of $Ax = 0$, what is the value of $\|Ax\|^2 = x^T A^T A x$?

- ☐ A 0
- ☒ C The smallest eigenvalue of $A^T A$ ✓
- ☐ B 1
- ☐ D The largest eigenvalue of $A^T A$

Answer C $x^T A^T A x = \lambda x^T x = \lambda$, so **the cost equals the eigenvalue**, and the optimum uses the smallest one. The cost is 0 only if all points satisfy the equation exactly (no noise). [Theory §4](#)

Q27 MEDIUM Inhomogeneous systems · plane fitting

When fitting the plane $z = ax + by + c$ by least squares, what is the i -th row of the coefficient matrix (unknowns $[a \ b \ c]^T$, right side z_i)?

- ☒ A $[x_i \ y_i \ 1]$ ✓
- ☐ B $[x_i \ y_i \ z_i \ 1]$
- ☐ C $[x_i \ 1]$
- ☐ D $[x_i^2 \ y_i^2 \ 1]$

Answer A Each point gives $z_i = ax_i + by_i + c$ (matrix.pdf §4.2). $[x_i \ y_i \ 1]$ is the row of the *homogeneous* implicit plane $ax + by + cz + d = 0$. [Theory §2](#)

Bonus Q28–Q33 • 6 questions

Lower probability given last week's format (numbers and concepts), but useful for understanding.

Q28 BONUS RANSAC • iterations (numeric)

RANSAC line fitting with confidence $p = 0.99$ and 50% inliers. About how many iterations are needed?

A ≈ 5

B ≈ 17 ✓

C ≈ 35

D ≈ 146

Answer B $s = 2: \ln(0.01) / \ln(1 - 0.25) = -4.605 / -0.288 \approx 16.0$, which rounds up to **17**. 35 would be for $s = 3$, and 146 for $s = 5$. **Theory §7**

Q29 BONUS Point-plane distance (numeric)

What is the distance of the point $(1, 1, 1)$ from the plane $2x + y + 2z + 4 = 0$?

A 1

B 9

C 3 ✓

D $9/\sqrt{13}$

Answer C $|2 + 1 + 2 + 4| / \sqrt{4 + 1 + 4} = 9/3 = 3$. The option $9/\sqrt{13}$ is the classic mistake of putting $d = 4$ under the root. **Theory §5**

Q30 BONUS RANSAC • concept

Why does RANSAC use a **minimal** sample for model estimation?

A Least squares cannot be applied to more points

B The probability of an all-inlier sample, w^s , drops exponentially with s ✓

C It makes the final model more accurate

D It removes noise from the inliers

Answer B A smaller s means a higher chance that a sample is outlier-free, so fewer iterations are needed. Accuracy comes afterwards, from the least-squares refit on all inliers. **Theory §7**

Q31 BONUS RANSAC • adaptive

In adaptive RANSAC, what happens to the planned number of iterations when a model with more inliers is found?

A It increases

B It decreases ✓

C It stays constant

D It is reset to the maximum

Answer B The estimated w increases, so w^s increases and k decreases. The lab printed $256 \rightarrow 125 \rightarrow 10$. **Theory §7**

Q32 BONUS RANSAC · concept

What is usually done after the RANSAC loop finishes?

A Refit the model with least squares using all inliers of the best model ✓

B Return the minimal sample of the best model unchanged

C Automatically fit a second model to the outliers

D Double the threshold and rerun

Answer A The model from the minimal sample is noisy. The final model is refit on all inliers, for example with centroid + smallest-eigenvalue eigenvector for a line or plane. Fitting another model to the remaining points is *sequential* multi-model RANSAC, and it isn't automatic. [Theory §6-7](#)

Q33 BONUS Lagrange multipliers

In a Lagrangian $J = f(x) - \lambda(g(x) - c)$, what does setting $\partial J / \partial \lambda = 0$ yield?

A The gradient of f

B The constraint $g(x) = c$ ✓

C $\lambda = 0$

D The Hessian of f

Answer B $\partial J / \partial \lambda = -(g(x) - c) = 0$ gives back the constraint. $\partial J / \partial x = 0$ gives the parallel-gradient condition $\nabla f = \lambda \nabla g$. [Theory §3](#)

One-page cheat sheet

Gradients

$$\partial(x^T x) = 2x$$

$$\partial(a^T x) = a$$

$$\partial(x^T A x) = (A^T + A)x$$

$$\partial(x^T B x) = 2Bx, \quad B = B^T$$

Inhomogeneous $Ax = b, m > n$

$$\min \|Ax - b\|^2$$

$$\text{normal eq. } A^T A x = A^T b$$

$$x = (A^T A)^{-1} A^T b = A^+ b$$

examples: $y = mx + b, z = ax + by + c$, circle, sphere

Homogeneous $Ax = 0$

$$\text{constraint } x^T x = 1$$

$$J = x^T A^T A x - \lambda(x^T x - 1)$$

$$A^T A x = \lambda x, \text{ cost} = \lambda$$

smallest-eigenvalue eigenvector of $A^T A$

= **last column** of V in $A = USV^T$

Lagrange

$$J = f(x) - \lambda(g(x) - c)$$

$$\partial_x J = 0 \Rightarrow \nabla f = \lambda \nabla g$$

$$\partial_\lambda J = 0 \Rightarrow g(x) = c$$

Distances

$$d = \frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\sqrt{\mathbf{l}^T \mathbf{l}}}, \text{ and if } \|\mathbf{l}\| = 1: d = |\mathbf{l}^T \mathbf{x}_0 + m|$$

$$\text{line: } \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} \cdot \text{plane: } \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

circle/sphere: $\|\mathbf{p} - \mathbf{c}\| - R$ · constant term never under the root

Optimal line/plane fit

passes through the **centroid** $\bar{\mathbf{x}} = \frac{1}{N} \sum \mathbf{x}_i$

normal = **smallest-eigenvalue** eigenvector of centered $X^T X$

$$m = -\mathbf{l}^T \bar{\mathbf{x}}$$

2D line direction = eigenvector of the largest eigenvalue

RANSAC

sample s points $\rightarrow$ fit $\rightarrow$ distances $\rightarrow$ inliers ($<$ threshold) $\rightarrow$ keep best $\rightarrow$ update $k \rightarrow$ finally refit with LSQ on all inliers

$$k = \left\lceil \frac{\log(1-p)}{\log(1-w^s)} \right\rceil \cdot P(\text{all-inlier sample}) = w^s \cdot \text{independent of } N, \text{ grows exponentially in } s$$

$$p = 0.99, w = 0.5: s = 2 \rightarrow 17, 3 \rightarrow 35, 4 \rightarrow 72, 5 \rightarrow 146$$

Minimal points

2D line **2** plane **3** circle **3** sphere **4** ellipse **5** parabola **3** 3D line **2** 3D circle **3** cylinder **5**

rule: DoF $\div$ constraints per point · implicit (homogeneous) models: DoF = #coefficients - 1